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Off Target Primer Binding Prevention Via Reverse Transcription Primer Design

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https://github.com/lerrocha/RT-Primer-Tool-.git

Abstract

Reverse Transcription (RT) primers are used in RNA sequencing to convert RNA into complementary DNA (cDNA). With that said RT primers can bind nonspecifically to unintended RNA sequences, especially ribosomal RNA (rRNA), resulting in inaccurate or contaminated data interpretation. In previous studies RT mispriming has occurred with minimal sequence complementary, especially near 3' end of the primer [1]. In this project, we developed a computational tool to evaluate RT primer specificity and reduce off-target binding. This program analyzes primer sequences by comparing binding similarities to both intended target genes and rRNA sequences. This tool evaluates primers intended for C. elegans genes xbp-1 and ets-4, they were experimentally observed to produce rRNA contamination during RT. The program calculates best binding percentage between primer and target sequence, including rRNA by generating single-base mutant primers to provide alternative sequences that maintain strong target binding while reducing rRNA homology [5].

Introduction

The grad student we were talking to was trying to use 5'RACE (rapid amplification of cDNA ends) to enrich for endogenously cleaved target RNA *xbp-1* and *ets-4* in *C. elegans* total RNA, but just ended up capturing contaminating rRNA [2-4]. This happened because the *xbp-1* RT/cDNA PCR reverse primer (JA-MV-78) bound to rRNA instead of the target RNA (*xbp-1)*[5]. The way that 5'RACE works is you ligate an adapter (containing a 3'hydroxyl) to cleaved RNAs (containing a 5' monophosphate), then make cDNA via reverse transcription (RT) with an RT primer that binds to a specific targeted locus (in this case, *xbp-1* or *ets-4*), then perform cDNA PCR with a forward primer that binds to the ligated-adapter sequence and the same reverse primers used in RT to amplify that particular target RNA [6]. This way you selectively amplify cleaved, adapter-ligated RNAs from a particular genomic locus into complementary dsDNA.

Our Solution

The grad student asked us to design a tool to help design RT primers for xbp-1 and ets-4 that minimally binds to contaminating rRNA to improve on-target amplification, such that the primer has minimal homology to the rRNA [2-4]. They also requested that the code gives suggestions for any mutations of the inputted primers [5]. Ideally, we could just test every permutation of primer of a certain length that would be below a specified percent match to the rRNA, but with the ideal primer lengths being between 18-30 base pairs, this is not feasible. Instead, the grad student requested that we have the tool analyze the inputted primer and report back on how strongly the primer would bond to the target and how strongly it would bind to the rRNA (Fig 1-2). From there, the code also gives suggestions on whether any point mutations to the primer will lead to less rRNA binding.

```
Length of xbp1 sequence = 1550
Length of ets4 sequence = 3732
Length of rrna sequence = 3509

xbp-1 reverse complement = CGTTGCTGGCTACATTGAGG
ets-4 reverse complement = TCCTTCCCCTGTTTACTAGCCT

Does xbp1 contain xbp1 primer reverse compliment? True
Does ets4 contain ets4 primer reverse compliment? True


============================================================
Analyzing primer: xbp-1 primer
Primer sequence: CCTCAATGTAGCCAGCAACG
============================================================
Reverse complement used for search: CGTTGCTGGCTACATTGAGG
Best on-target sequence match to xbp-1 = 100.0 %
Best sequence match to rRNA (off-target risk) = 60.0 %
This primer fails the off-target rule.

Suggested alternative primers:
Primer: TCTCAATGTAGCCAGCAACG Target match: 95.0 % rRNA Match: 60.0 %
Primer: GCTCAATGTAGCCAGCAACG Target match: 95.0 % rRNA Match: 60.0 %
Primer: CATCAATGTAGCCAGCAACG Target match: 95.0 % rRNA Match: 60.0 %
Primer: CGTCAATGTAGCCAGCAACG Target match: 95.0 % rRNA Match: 60.0 %
Primer: CCACAATGTAGCCAGCAACG Target match: 95.0 % rRNA Match: 60.0 %
Primer: CCTAAATGTAGCCAGCAACG Target match: 95.0 % rRNA Match: 60.0 %
Primer: CCTTAATGTAGCCAGCAACG Target match: 95.0 % rRNA Match: 60.0 %
Primer: CCTCTATGTAGCCAGCAACG Target match: 95.0 % rRNA Match: 60.0 %
Primer: CCTCGATGTAGCCAGCAACG Target match: 95.0 % rRNA Match: 60.0 %
Primer: CCTCATTGTAGCCAGCAACG Target match: 95.0 % rRNA Match: 60.0 %
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Fig. 1 Analysis of a proposed XBP-1 primer and possible alternatives

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Best alternative found:
Primer: TCTCAATGTAGCCAGCAACG
Target match: 95.0 %
rRNA match =  60.0 %
GC content: 50.0% GC
This suggestion improves the primer but fails the 30% off-target rule.

============================================================
Analyzing primer: ets-4 primer
Primer sequence: AGGCTAGTAAACAGGGGAAGGA
============================================================
Reverse complement used for search: TCCTTCCCCTGTTTACTAGCCT
Best on-target sequence match to ets-4 = 100.0 %
Best sequence match to rRNA (off-target risk) = 59.09 %
This primer fails the off-target rule.

Suggested alternative primers:
Primer: AGCCTAGTAAACAGGGGAAGGA Target match: 95.45 % rRNA Match: 54.55 %
Primer: AGGCTAGTAACCAGGGGAAGGA Target match: 95.45 % rRNA Match: 54.55 %
Primer: AGGCTAGTAAGCAGGGGAAGGA Target match: 95.45 % rRNA Match: 54.55 %
Primer: AGGCTAGTAAACAGGGGACGGA Target match: 95.45 % rRNA Match: 54.55 %
Primer: AGGCTAGTAAACAGGGGAAGCA Target match: 95.45 % rRNA Match: 54.55 %
Primer: AGGCTAGTAAACAGGGGAAGGT Target match: 95.45 % rRNA Match: 54.55 %
Primer: TGGCTAGTAAACAGGGGAAGGA Target match: 95.45 % rRNA Match: 59.09 %
Primer: CGGCTAGTAAACAGGGGAAGGA Target match: 95.45 % rRNA Match: 59.09 %
Primer: GGGCTAGTAAACAGGGGAAGGA Target match: 95.45 % rRNA Match: 59.09 %
Primer: ACGCTAGTAAACAGGGGAAGGA Target match: 95.45 % rRNA Match: 59.09 %

Best alternative found:
Primer: AGCCTAGTAAACAGGGGAAGGA
Target match: 95.45 %
rRNA match =  54.55 %
GC content: 50.0% GC
This suggestion improves the primer but fails the 30% off-target rule.
```

Fig. 2 Analysis of a proposed ets-4 primer and possible alternatives

Discussion

The results of this project show how computational analysis can help evaluate primer specificity and identify potential off-target binding. Since rRNA is highly abundant in cellular RNA samples, weak primer homology can lead to off-target amplification. Detecting these interactions before conducting experiments can save both time and resources [5-6]. The program evaluates sequence similarity between primers and potential target sequences, providing an estimated measure of binding. However, the current program still has several limitations where it only currently checks sequence similarity. It does not take into consideration other factors that influence primer performance such as melting temperature, primer secondary structure, or RNA accessibility. In addition, the program only conducts single-base mutations which could limit the range of alternative primer designs that can be discovered. These features that our program lacks highlights areas that could be set as future improvements for this tool.

Conclusion

In summary, our code's primary function is primer analysis. Our program takes in an input of target sequences, a primer, and an rRNA sequence [2-5]. It then processes these inputs and returns multiple items. One such output is the strength that this original primer sequence would bind to the target, presented in the form of a percent match and a percent GC match. It also returns how at risk the primer is of mistakenly binding to the rRNA sequence, also presented in the form of percent match and percent GC match. The last output details how any replacement mutations could affect primer binding strength and the likelihood of the primer binding to the rRNA in a desirable manner. This is presented in the form of a list of all the mutated primer sequences, which when compared to the original primer sequence, have reduced rRNA contamination. This program provides a method of reducing rRNA contamination and we hope that in the future this can contribute to furthering the automation of primer design and the development of tools to improve existing data.

References

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